

GUANTANAMO BAY NAS, Cuba

WMO: 783670

Lat: 19.906N

Lon: 75.207W

Elev: 56

StdP: 14.67

Time Zone: -5.00 (CUB)

Period: 94-19

WBAN: 11706

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	65.3	66.8	55.1	64.8	71.6	57.2	69.9	72.4	25.2	79.3	23.0	81.1	9.3	340	0.427

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.7	93.1	79.2	91.9	79.0	91.2	78.8	82.3	89.1	81.5	88.5	80.7	87.9	16.6	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.2	157.0	86.6	79.2	151.7	86.0	78.5	148.5	85.7	46.1	89.4	45.1	88.5	44.3	88.0	87.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.4	21.1	18.9	62.4	96.0	2.7	2.3	60.5	97.7	58.9	99.1	57.4	100.4	55.4	102.1
			57.7	84.1	3.4	2.2	55.2	85.6	53.3	86.9	51.4	88.2	49.0	89.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	80.4	76.7	76.8	77.6	79.2	80.9	82.5	83.7	83.8	82.9	81.7	80.0	78.1
	DBStd	3.22	2.28	2.15	2.23	1.92	2.00	1.97	1.70	1.68	1.71	1.75	2.26	2.35
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11080	829	751	857	875	957	975	1046	1047	988	983	900	872
	CDD65	5605	364	331	392	425	492	525	581	582	538	518	450	407
	CDH74	57525	2734	2677	3341	4066	5174	6244	7270	7216	6217	5321	4019	3246
	CDH80	21536	739	696	908	1277	1865	2566	3277	3220	2588	1966	1402	1032
Wind	WSAvg	9.2	9.6	9.9	9.9	9.8	8.9	8.5	9.5	9.3	8.6	8.5	9.0	9.2
Precipitation	PrecAvg	22.3	1.1	0.8	1.2	1.2	3.3	2.0	0.9	2.0	2.9	4.8	1.5	1.1
	PrecMax	44.5	10.4	4.4	5.5	5.4	18.8	8.0	4.1	5.7	18.0	19.8	5.4	6.9
	PrecMin	9.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.1	0.0
	PrecStd	9.0	1.8	1.0	1.3	1.4	3.9	1.9	0.8	1.4	4.0	4.6	1.4	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.1	88.0	88.6	89.4	91.3	92.8	94.2	94.4	93.9	93.0	91.8	89.6
		MCWB	73.8	74.0	74.8	76.1	77.6	78.1	79.2	80.1	80.1	79.7	78.3	75.8
	2%	DB	86.5	86.4	87.1	88.0	89.5	91.4	92.9	92.8	92.0	90.0	89.5	87.9
		MCWB	73.9	73.6	74.6	75.8	77.0	78.2	78.5	79.9	79.8	78.5	77.6	75.5
	5%	DB	85.4	85.2	85.7	86.8	88.1	89.8	91.6	91.6	90.0	89.2	88.0	86.5
		MCWB	73.8	73.4	74.3	75.5	77.0	78.2	78.5	79.6	79.1	78.3	77.0	75.2
	10%	DB	84.2	84.2	84.4	85.8	87.4	88.7	89.9	89.9	89.2	87.9	86.5	85.2
		MCWB	73.4	73.3	73.9	75.1	76.9	78.1	78.3	78.9	78.9	78.1	76.4	74.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.1	77.2	80.6	80.5	81.9	82.5	83.1	83.1	83.1	82.4	81.0	79.5
		MCDB	84.0	84.1	82.6	85.6	86.8	88.7	90.2	90.6	90.0	88.7	87.2	85.3
	2%	WB	76.4	76.0	77.2	78.9	80.5	81.1	81.6	81.8	81.7	80.9	79.7	77.8
		MCDB	83.3	83.5	84.2	85.0	86.0	88.0	89.3	89.8	88.8	87.3	86.5	84.8
	5%	WB	75.4	75.2	76.1	77.5	79.1	80.1	80.5	80.9	80.7	79.9	78.6	76.7
		MCDB	82.8	83.0	83.4	84.2	85.4	87.4	88.6	89.1	88.0	86.6	86.0	84.3
	10%	WB	74.4	74.3	75.1	76.4	78.0	79.2	79.6	80.0	79.8	79.0	77.5	75.7
		MCDB	82.0	82.3	82.7	83.5	84.8	86.9	88.2	88.4	87.3	86.0	84.8	83.3
Mean Daily Temperature Range	5% DB	MDBR	15.6	15.5	14.6	13.9	13.1	12.7	13.7	13.7	13.7	13.4	14.5	15.2
		MCDBR	16.6	16.3	15.4	14.6	13.9	13.9	15.0	14.9	14.6	14.9	15.5	16.2
		MCWBR	8.5	8.6	7.9	7.6	7.2	7.2	7.4	7.4	7.0	7.0	7.4	7.9
	5% WB	MCDBR	15.3	15.3	14.1	13.6	12.7	13.1	13.9	14.0	14.0	13.9	14.5	15.3
		MCWBR	9.0	9.3	8.7	7.6	7.0	7.2	7.7	7.6	7.1	7.1	7.6	8.3
Clear-Sky Solar Irradiance	taub	0.376	0.383	0.392	0.422	0.450	0.493	0.502	0.468	0.428	0.403	0.390	0.386	
	taud	2.529	2.498	2.469	2.397	2.348	2.214	2.174	2.305	2.437	2.513	2.514	2.513	
	Ebn at Noon	286	290	291	281	270	256	254	265	276	280	279	279	
	Edh at Noon	31	34	36	39	40	46	48	42	36	33	31	31	
All-Sky Solar Radiation	RadAvg	1441	1716	1919	2034	1941	1958	2049	1982	1836	1570	1432	1361	
	RadStd	92	91	96	105	111	122	88	71	109	132	101	79	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	+2.99	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	N/A	N/A	+2.99	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page